

A PROJECT TITLE ON
Fraudulent Transaction Clustering Using DBSCAN-Based Anomaly Detection

project report submitted in partial fulfilment of the requirements for the Course

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DEPARTMENT OF **Computer Science and Engineering**

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1. Abstract:

Fraudulent financial transactions have become one of the biggest challenges for banks and digital payment platforms. Traditional fraud detection methods often require labeled datasets and predefined rules, making it difficult to identify new or evolving fraud patterns. This project proposes a fraud detection system using the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm, an unsupervised machine learning technique capable of detecting anomalies without prior labeling.

The system analyzes transaction attributes such as transaction amount, time, location, account behavior, and transaction frequency to identify clusters of normal transactions while treating isolated transactions as anomalies. Since fraudulent transactions usually differ significantly from regular customer behavior, DBSCAN effectively identifies these unusual patterns as potential fraud.

The proposed system improves fraud detection accuracy, reduces manual investigation effort, and provides an intelligent solution for identifying suspicious transactions in real time. The project demonstrates how density-based clustering can enhance financial security by automatically recognizing abnormal transaction behavior.

2. INTRODUCTION:

With the rapid growth of online banking, mobile payments, e-commerce, and digital financial services, the number of electronic transactions has increased dramatically. Along with this growth, fraudulent financial activities such as credit card fraud, identity theft, unauthorized transactions, and account hacking have also increased.

Detecting fraudulent transactions is a major challenge because fraudsters continuously change their methods to avoid detection. Traditional rule-based fraud detection systems depend on predefined rules and historical fraud records, making them less effective against new fraud patterns. To overcome these limitations, machine learning techniques have become popular in fraud detection. Among these methods, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is an effective unsupervised clustering algorithm capable of identifying abnormal transactions without requiring labeled training data.

This project develops a fraud detection system that groups similar transactions into clusters while identifying isolated transactions as anomalies. These anomalies are treated as fraudulent activities, allowing financial institutions to quickly investigate suspicious transactions and minimize financial losses.

3. SOFTWARE REQUIREMENTS:

- **Python:** Used to build the complete fraud detection application.
- **VS-Code:** Provides an environment to write and execute Python code.
- **Pandas:** Reads and organizes transaction data from CSV files.
- **NumPy:** Performs fast mathematical operations required for data processing.
- **Scikit-learn:** Implements the DBSCAN algorithm to identify fraudulent transactions.
- **Matplotlib:** Displays graphs that help visualize clusters and anomalies.
- **CSV Dataset:** Acts as the input data source containing transaction records.
- **Operating System:** Supports running the software and required libraries.